

HYPOTHYROIDISM

- ↓sed secretion of thyroid hormone.
- Hypothyroidism leads to:
 - **Myxedema** in adults
 - **Cretinism** in children

MYXEDEMA

- Hypothyroidism in adults, characterised by **generalised edematous appearance**.

Causes for Myxedema:

- Due to disease of thyroid gland, genetic disorder, or iodine deficiency.
- Also due to deficiency of TSH & TRH.
- **Specific causes:**
 - a. Common cause is autoimmune disease called **Hashimoto's thyroiditis**.
 - b. In most patients, it starts with glandular inflammation called **thyroiditis** (*caused by autoimmune antibodies*).

Signs and Symptoms:

- **Edematous appearance throughout body:**
 - a. Swelling of face
 - b. Bagginess under eyes
 - c. **Non-pitting type of edema** (*does not make pits when pressed*):
 - ↓
 - Becoz of accumulation of proteins with **hyaluronic acid** and **chondroitin sulfate**.
 - d. Atherosclerosis (*becoz of ↑sed plasma level of cholesterol*)
 - e. Anemia
 - f. Fatigue & muscular sluggishness
 - g. Menorrhagia & polymenorrhea
 - h. ↑sed in body weight
 - i. Scaliness of skin

- j. Frog-like husky voice

CRETINISM

- **Hypothyroidism in children**, characterised by **stunted growth**.

Causes of Cretinism:

- Due to **congenital absence of thyroid gland**, genetic disorder, or lack of iodine in diet.

Features of Cretinism:

- a. **Few weeks after birth**: Baby starts developing signs like sluggish movements and **croaking sound (crying)**.
- b. **Stunted growth** with bloated body.
- c. **Tongue becomes so big** that it hangs down with dripping of saliva.
- d. **Big tongue obstructs** swallowing and breathing.
- e. **Tongue produces guttural breathing** that may sometimes choke the baby.

Cretin vs Dwarf

CRETINISM	DWARFISM
• Mental retardation	• Development of nervous system is normal
• Different parts of body are disproportionate	• Parts of body are proportionate
• Reproductive function affected	• Reproductive function normal

GOITER

- **Enlargement of thyroid gland**.
- Occurs in both **hypothyroidism** and **hyperthyroidism**.

1. TOXIC GOITER – GOITER IN HYPERTHYROIDISM

- Enlargement of thyroid gland with **↑sed secretion of thyroid hormones**, caused by **thyroid tumor**.

2. NON-TOXIC GOITER – GOITER IN HYPOTHYROIDISM

- Enlargement of thyroid gland **without ↑se in hormone secretion**.
- Also called **hypothyroid goiter**.

Classification of Non-Toxic Hypothyroid Goiter:

- a. Endemic colloid goiter
- b. Idiopathic non-toxic goiter

A. Endemic Colloid Goiter

- The non-toxic goiter caused by **iodine deficiency**.
- Also called **iodine deficiency goiter**:
 - ↓ *when intake* < 50 μg/day
- **Pathophysiology**:
 - No formation of hormones (*lack of iodine*)
 - Thyroglobulin remains same (*no hormone present*)
 - ↓
 - Thyroglobulin accumulated in follicles of gland
 - ↓
 - ↑ses size of gland
- **Endemic area**: Very common in **Kashmir valley** (*soil does not contain iodine → foodstuffs do not contain I*).

B. Idiopathic Non-Toxic Goiter

- Goiter due to **unknown cause**.
- Enlargement of thyroid gland occurs **even without iodine deficiency**.
- **Etiology:**
 - May be due to thyroiditis and **deficiency of enzymes** such as *peroxidase, iodinase, & deiodinase (required for thyroid hormone synthesis)*.
 - Some foodstuffs contain goitrogenic substance:
 - ↓
 - **Goitrin** (*contains antithyroid substance like propylthiouracil*).

GOITROGENS

(Suppress synthesis of thyroid hormones)

- TSH secretion ↑ ~~ses~~ → resulting in enlargement of gland.
- **Found in vegetables:** Turnips & cabbage (*soybean also*).
- **Goitrogens become active only during low iodine intake.**